لجنة الطب البشري
فكرة تحيا على نبض قلوبكم

(Quiz #5&6)

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**Kindly, write true (T) beside the correct sentence ,& write false (F)beside the wrong sentence**

- ( F ) : **cellular respiration is aerobic process** only
- ( F ) : **the total energy released from fermentation is 20 ATP**
- ( T ) : **the location of cellular perspiration is in mitochondria**
- ( T ) : **co2 is released from exhalation process**
- ( F ) : **dehydrogenase activity was detected by using dye as iodine**
- T ( ) : **we used yeast as source of dehydrogenase**
- ( F ) : **the enzyme used in reactions as catalyst and consume during the reaction**
- ( F ) : **in dehydrogenase activity , we used glucose as an acceptor**
- ( F ) : **fermentation more efficiently than respiration and yield more ATP**
- ( F ) : **the optimum temperature for catalase in plant is37 C**
- T ( ) : **the catalase active in acid condition**

**BA[OH]2+CO2** ⟶ $BaCO_3$ + $H_2O$

**H2O2 +CATALASE** ⟶ $2H_2O$ + $O_2$

| | # co2 | Yield of ATP |
|---|---|---|
| **RESPIRATION** | $6CO_2$ | 34 - 36 ATP |
| **FERMANTATION** | $2CO_2$ | 2 ATP |

Yeast

- Dehydrogenase + glucose + methylene Blue → $\underbrace{\text{leuko}}_{\text{white}}$ methylen

- yeast cells are Facultative anaerobic They metabolize sugars To produce 1- ~~methanol~~ $6H_2O$ 2- $6CO_2$ ~~and~~ with $O_2$ and without $O_2$ They produce 1- $CO_2$ 2- ~~methanol~~ ethanol by metabolize pyruvate and detected The reaction by change pH indicator by using phenol red

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15

## LABORATORY REPORT

### Cellular Respiration and Fermentation Laboratory

[redacted]

### A. Carbon Dioxide ($CO_2$) Liberation exercise:

1- Record the color and the changes in the exhale and inhale bottles.

Exhale : white Inhale : colorless

2- Explain your observation and Write the chemical equation of the reaction.

$Ba(OH)_2 + O_2 \longrightarrow$ No reaction Inhale

$Ba(OH)_2 + CO_2 \longrightarrow BaCO_3 + H_2O$ Exhale

this material reacts with $CO_2$ so the change will be white Exhaling

### B. Water liberation

1- Record what happened on the surface of the glass test tube after you exhale in it several times, explain your observation.

consentration of $H_2O$ on the glass exhale $\longrightarrow CO_2 + H_2O$

### C. Heat of Respiration

1- Record the temperature in each chamber at the end of your lab and explain the differences in the temperature reading.

Dead : 18 no heat because the denaturation of enzymes so no ATP produced

Live : 23 there will be heat because of the formation of ATP by cellular respiration.

### D. Dehydrogenase activity

1- Record the results of your exercise by filling the table below:

| Tube # | Contents | Time for methylene blue to go colourless |
|---|---|---|
| 1 | Glucose + dye + yeast | 15 min<br>+ve |
| 2 | Glucose + dye + boiled yeast | 0<br>-ve |
| 3 | Glucose + dye + buffer | 0<br>-ve |

E. Anaerobic Respiration of yeast cells



1- Record the results of your exercise by filling the table below:

| Tube # | Contents | Final Air space length (mm) in the vial | Color change |
|---|---|---|---|
| 1 | Water + yeast | 0 | red |
| 2 | Glucose + yeast | 60 mm | Yellow |
| 3 | Sucrose + yeast | 0 | red |
| 4 | Starch+ yeast | 0 | red |

2- Explain what gas was accumulated in the vials? Why the color of the dye changed?.

$CO_2$/ because the dye reacted and changed color in acidic solution to yellow

**GOOD LUCK**

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